Approximation of areas

1 Use the trapezoidal rule to find an approximation for each integral:

a $\int_2^5 3x^2\,dx$ **b** $\int_0^2 (2x^3 - 1)\,dx$ **c** $\int_2^4 dx$

2 Find an approximation to $\int_0^1 x^3\,dx$ using the trapezoidal rule with:

a 1 subinterval **b** 2 subintervals

3 Given the table of values, find the approximate value of each definite integral:

a $\int_0^8 f(t)\,dt$

t	0	2	4	6	8
$f(t)$	3	7	11	9	3

b $\int_0^{40} f(x)\,dx$

x	0	10	20	30	40
$f(x)$	350	410	435	450	460

c $\int_{-4}^2 f(x)\,dx$

x	-4	-3	-2	-1	0	1	2
$f(x)$	0	4	5	3	10	11	2

4 Use the trapezoidal rule with 2 subintervals to find an approximation for the following to three decimal places:

a $\int_3^4 \ln 2x\,dx$ **b** $\int_0^6 \frac{dx}{x+3}$

5 Find an approximation to the following integrals, rounding your answers to three decimal places:

a $\int_2^3 e^{2x}\,dx$ using 3 subintervals

b $\int_0^2 (4x^2 - 16)\,dx$ using 4 subintervals

c $\int_0^1 \sqrt{2x}\,dx$ using 5 subintervals

d $\int_1^3 \frac{dx}{x^3}$ using 4 subintervals

e $\int_2^5 \frac{2}{3x-4}\,dx$ using 6 subintervals

6

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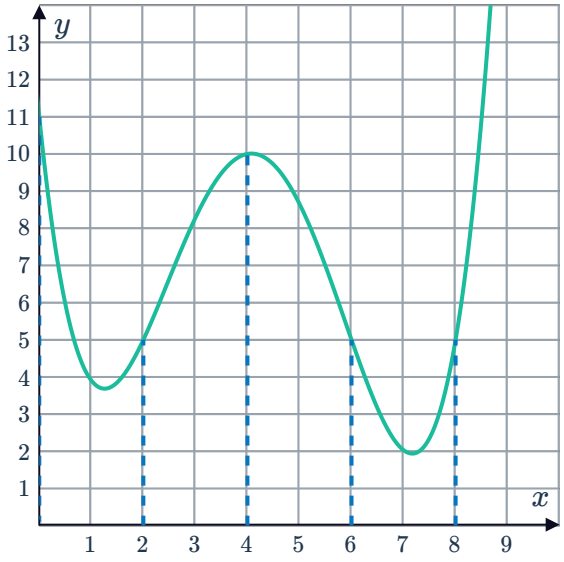
Use one application of the trapezoidal rule to approximate $\int_2^3 \frac{e^x}{x}\,dx$ to one decimal place.

7 Use three applications of the trapezoidal rule to approximate $\int_0^9 e^{-x^2}\,dx$ to one decimal place.

8 Approximate $\int_0^8 8x\,dx$ by using four rectangles of equal width whose heights are the values of the function at the midpoint of each rectangle.

9 Approximate $\int_1^5 \frac{1}{x}\,dx$ by using four rectangles of equal width whose heights are the values of the function at the right endpoint of each rectangle.

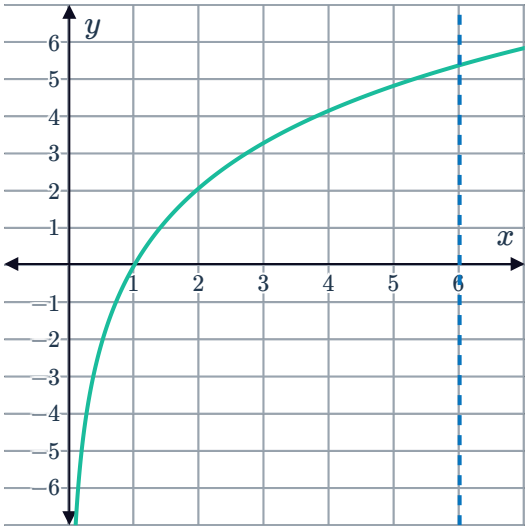
10 In the following graph, the interval $[0, 8]$ is partitioned into four subintervals $[0, 2]$, $[2, 4]$, $[4, 6]$, and $[6, 8]$:



a Approximate the area A using rectangles for each subinterval whose heights are equal to the function values of the left side of the subintervals.

b Approximate the area A using rectangles for each subinterval whose heights are equal to the function values of the right side of the subintervals.

11 The function $y = 3\ln x$ has been graphed: Use two applications of the trapezoidal rule to approximate the area bound by the curve, the x-axis and and $x = 6$. Round your answer to one decimal place.



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12 Consider the function $y = e^{x^2}$.

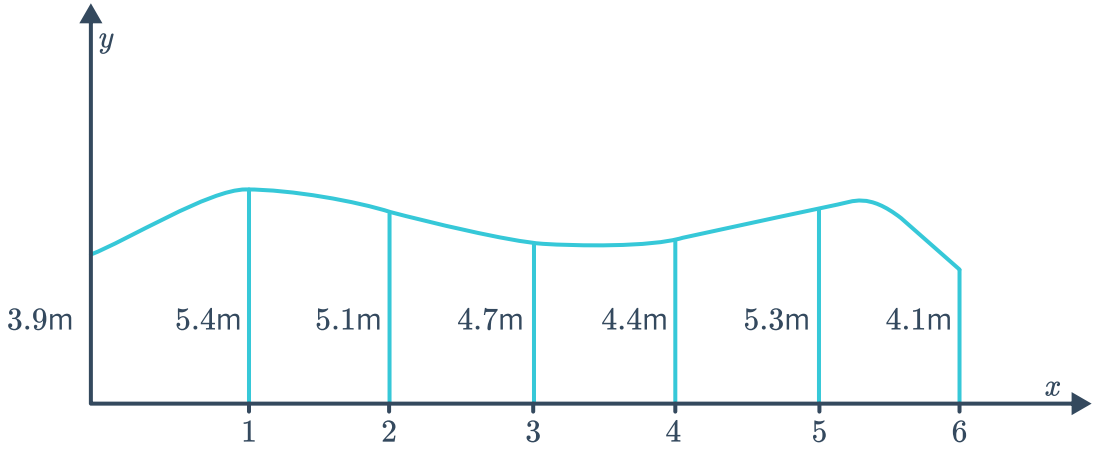
a Determine y''.

b Using two applications of the trapezoidal rule, approximate the area bound by the curve and the x-axis between $x = 0$ and $x = 2$. Round your answer to one decimal place.

c Is the approximation given by the trapezoidal rule an underestimate or an overestimate of the actual area? Explain your answer.

13 Consider the integral $\int_1^5 (\ln x + 4)\, dx$.
Approximate the area bounded by the curve, the x-axis, $x = 1$ and $x = 5$ using two applications of the trapezoidal rule. Round your answer to two decimal places.

14 Use the trapezoidal rule to find the approximate area of the irregular figure below:



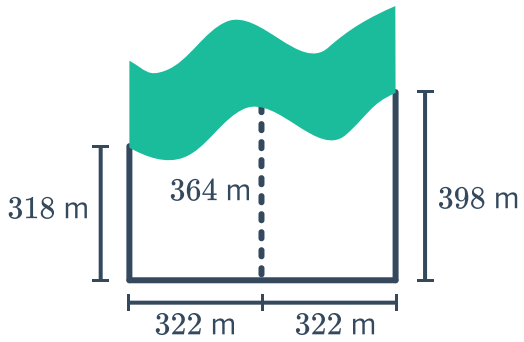
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Applications

15 The following piece of land has straight boundaries on the east, west and south borders and is bounded by a creek to the north. The land has been divided into two sections so we can use the trapezoidal rule to approximate the area:

a Find the approximate area of the piece of land by using two applications of the trapezoidal rule.

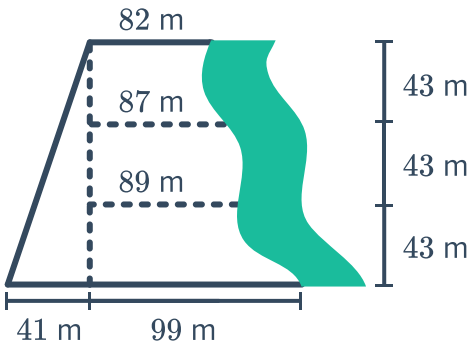
b During a heavy storm, $35.2\ \text{mm}$ of rain fell. Find the volume of water that falls on this land. Round your answer to the nearest cubic metre.



16 A surveyor made the following diagram with measurements for a property she was mapping out. On the west side of the property is a river:

a Find the approximate total area of the property by using three applications of the trapezoidal rule.

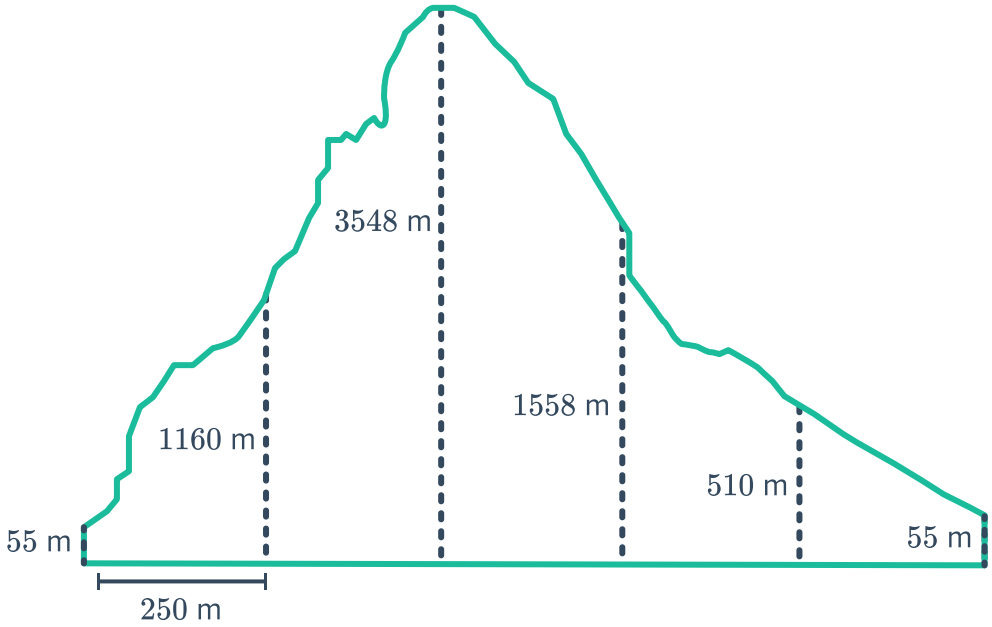
b The average weekly rainfall is $34\ \text{mm}$. Calculate the total volume of water that falls on the land in cubic metres. Round your answer to two decimal places.



17 A river has its depths marked out at equal intervals of $9\ \text{m}$. The depths are $0, 12, 14, 17, 5$, and $0\ \text{m}$ respectively. Find the approximate area of the cross section of the river.

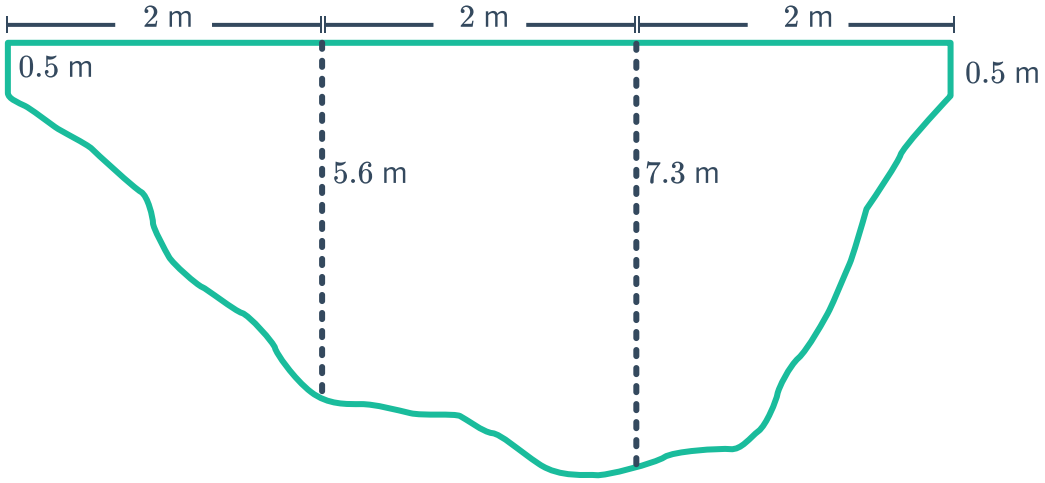
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18 The elevation values of a mountain are recorded at equal intervals of $250\ \text{m}$. The heights are shown in the diagram:



Find the approximate area of the cross section of the mountain.

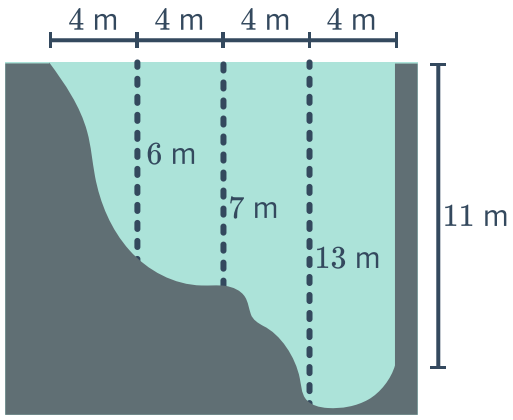
19 The diagram shows the cross-section of a river. The depths of the river are marked at 2-metre intervals:



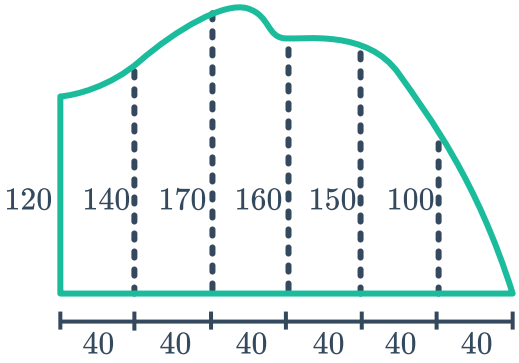
Using three applications of the trapezoidal rule, approximate the area of the cross-section of the river to one decimal place.

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20 Use four applications of the trapezoidal rule to approximate the area of the cross-section of the following river:



21 The following shape has measurements given in metres. Use the trapezoidal rule to find the area in hectares.



22 A garden is $49\ \text{m}$ long. At $7\ \text{m}$ intervals, the width of the garden was given by the following measurements:

$$0\ \text{m},\ 2.9\ \text{m},\ 5.2\ \text{m},\ 6.6\ \text{m},\ 5.6\ \text{m},\ 4.3\ \text{m},\ 3\ \text{m},\ 2.5\ \text{m}$$

Using the trapezoidal rule, approximate the area of the garden to two decimal places.

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